

Requirement provenance — where each of the 74 features came from

A mark means the source's published documentation evidenced this requirement. It does NOT mean the product has the feature, and a blank does NOT mean it lacks one.

The research behind the gap matrix was a handful of published pages per vendor — enough to establish what the market expects a key management system to do, nowhere near enough to compare products feature by feature. A capability comparison would need a hands-on evaluation of each product, and this is not one.

THA Thales CipherTrust Manager	FOR Fortanix Data Security Manager	ENT Entrust KeyControl	UTI Utimaco Enterprise Secure Key Manager	CLD AWS KMS / Azure Managed HSM / Google Cloud KMS	STD NIST SP 800-57, SP 800-152, OASIS KMIP 2.1 and 3.0	CAT Category baseline — no specific source named it
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CAT is the honest residue: requirements no specific source in that research named, present because they are ordinary practice for a networked service that handles secrets. A row carried only by CAT is the weakest-sourced kind in the table, and is marked in amber so it reads that way at a glance rather than hiding among the others.

Key lifecycle and cryptography

Requirement	THA	FOR	ENT	UTI	CLD	STD	CAT	Status
HSM-backed storage, keys never leave	●	●	●	●	●	●	-	Covered
Full lifecycle state machine	●	-	●	-	-	●	-	Covered
Cryptoperiod enforcement	-	-	●	-	●	●	-	Covered
Automatic rotation	●	-	●	●	●	-	-	Partial
Key lineage and versioning	●	-	-	-	●	-	-	Covered
Algorithm breadth	-	-	-	-	-	●	●	Partial
Post-quantum (ML-KEM, ML-DSA, SLH-DSA)	-	-	-	-	-	●	●	Gap
Hardware RNG	-	-	-	●	-	●	-	Covered
Key derivation	-	-	-	-	-	●	●	Covered
Key wrapping / secure import-export	●	-	-	-	●	●	-	Covered
Split knowledge / M-of-N shares	-	●	-	●	-	●	-	Partial
Formal key ceremony	-	-	-	-	-	●	●	Gap
Bulk and batch operations	-	-	-	-	-	●	●	Partial
Verifiable destruction	●	-	●	-	-	●	-	Covered

Protocol and ecosystem integration

Requirement	THA	FOR	ENT	UTI	CLD	STD	CAT	Status
KMIP server	●	-	●	●	-	●	-	Covered
KMIP interoperability certification	-	-	-	-	-	●	●	Gap
KMIP 3.0	-	-	-	-	-	●	-	Gap
PKCS#11 provider for applications	-	-	-	●	-	-	●	Gap
REST / JSON API	●	●	-	●	●	-	-	Gap
Cloud BYOK (AWS, Azure, GCP)	●	●	-	-	●	-	-	Gap
Cloud EKM / XKS / hold-your-own-key	-	●	-	-	●	-	-	Gap
Database TDE integration	●	-	●	-	-	-	-	Partial

Storage, backup and VM integrations	.	.	●	.	.	.	●	Partial
Secrets management	●	●	●	.	.	.	.	Partial
Tokenization / format-preserving encryption	●	●	.	.	.	.	.	Gap
Certificate lifecycle	.	.	.	.	.	.	●	Partial

Identity and access control

Requirement	THA	FOR	ENT	UTI	CLD	STD	CAT	Status
Per-identity authentication	●	.	.	●	.	.	●	Covered
Enterprise IdP — LDAP/AD, SAML, OIDC	.	.	.	●	.	.	●	Gap
mTLS certificate identity	.	.	.	.	.	.	●	Covered
API keys / service accounts	.	.	.	.	●	.	●	Gap
Role-based access control	●	.	.	●	.	.	●	Covered
Groups	.	.	.	.	.	.	●	Covered
Per-object access grants	●	.	.	.	.	.	●	Covered
Attribute or policy-based access	●	●	●	.	.	.	.	Gap
Dual control / quorum approval	.	●	.	●	.	.	.	Covered
Separation of duties	.	.	.	.	.	●	●	Partial
Multi-tenancy and namespaces	●	.	●	.	.	.	.	Gap
Per-tenant quotas	.	.	.	.	.	.	●	Gap
Connection and rate limiting	.	.	.	.	.	.	●	Gap

Audit, compliance and assurance

Requirement	THA	FOR	ENT	UTI	CLD	STD	CAT	Status
Tamper-evident audit log	●	.	.	●	.	●	.	Covered
External audit anchoring	.	.	.	.	.	.	●	Gap
SIEM integration	.	.	.	●	.	.	●	Gap
Retention and	.	.	.	.	.	●	●	Covered

What the columns add up to

Read this as a description of the research, not of the market. A source with few marks is one whose documentation I read less of, not a product with fewer features.

Source	Requirements evidenced	Share of 74	Kind
THA — Thales CipherTrust Manager	26	35 %	Vendor documentation
FOR — Fortanix Data Security Manager	11	14 %	Vendor documentation
ENT — Entrust KeyControl	18	24 %	Vendor documentation
UTI — Utimaco Enterprise Secure Key Manager	16	21 %	Vendor documentation
CLD — AWS KMS / Azure Managed HSM / Google Cloud KMS	11	14 %	Vendor documentation
STD — NIST SP 800-57, SP 800-152, OASIS KMIP 2.1 and 3.0	25	33 %	Standard
CAT — Category baseline — no specific source named it	45	60 %	Category baseline

Requirements carried only by the category baseline: 20 of 74

These are the rows to challenge first if the requirement list is ever disputed. Each is defensible as ordinary practice, but none was named by a vendor or a standard in the research behind the matrix.

- Certificate lifecycle
- mTLS certificate identity
- Groups
- Per-tenant quotas
- Connection and rate limiting
- External audit anchoring
- Crypto-agility reporting
- Enterprise database backend
- Zero-downtime certificate rotation
- Rolling upgrades
- Health and readiness probes
- Metrics
- Structured logging
- Configuration-driven deployment
- Administrative CLI
- Alerting
- TLS enforced by default
- Request bounds and DoS resistance
- Secrets kept out of configuration

- Hardened service

What would make this stronger

- A hands-on evaluation of one or two products, which would turn this provenance table into a genuine capability comparison.
- The vendors' administration guides rather than their marketing pages — the former say what a product does, the latter what it advertises.
- An analyst grid, which would add an independent view of the category rather than each vendor's own.